

# Discrete Math Scribe Notes: Functions & Relations

9/22/26

## 1 Cartesian Products

Let  $A = \{1, 2, 3\}$  and  $B = \{a, b, c\}$ .

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c), (3, a), (3, b), (3, c)\}$$

$$B \times A = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3), (c, 1), (c, 2), (c, 3)\}$$

**Note:** ordered pairs are compared coordinate by coordinate, so in general  $(a, 1) \neq (1, a)$  — this is exactly why  $A \times B$  and  $B \times A$  are different sets.

If  $|A| = x$  and  $|B| = y$ , then

$$|A \times B| = xy.$$

## 2 Functions: Domain, Codomain, Range

A function  $f : A \rightarrow B$  assigns to every  $x \in A$  a unique  $f(x) \in B$ :

$$\forall x \in A, f(x) \in B.$$

Here  $A$  is the **domain** and  $B$  is the **codomain**.

**Example.** Let  $A = \{a, b, c, d\}$ ,  $B = \{1, 2, 3, 4\}$ , with

$$f(a) = 2, \quad f(b) = 3, \quad f(c) = 1, \quad f(d) = 2.$$

- Domain:  $\{a, b, c, d\}$
- Codomain:  $\{1, 2, 3, 4\}$
- Range:  $\{1, 2, 3\}$  (note 4 is never hit, so it is in the codomain but not the range)

**Image and pre-image.** For  $y = f(x)$  with  $f : U \rightarrow V$  (where  $U \subseteq A$ ,  $V \subseteq B$ ),  $y$  is the **image** of  $x$ , and  $x$  is a **pre-image** of  $y$ .

Examples:  $f(x) = x^2$  with  $f : \mathbb{Z} \rightarrow \mathbb{N}$ ;  $f : \mathbb{R} \rightarrow \mathbb{R}$ .

In general,

$$\text{Range} = \{b \in B \mid \exists x \in A \text{ with } f(x) = b\}.$$

## 3 Injective and Surjective Functions

### 3.1 Injective (one-to-one)

$f : A \rightarrow B$  is **injective** if every element of  $A$  maps to a *unique* element of  $B$  (though not every element of  $B$  needs to be hit). Every function must send each element of  $A$  somewhere in  $B$  — injectivity is the extra requirement that no two inputs share an output:

$$\forall x, y \in A : f(x) = f(y) \iff x = y,$$

equivalently (the contrapositive form):

$$x \neq y \implies f(x) \neq f(y).$$

### 3.2 Surjective (onto)

$f : A \rightarrow B$  is **surjective** if every element of  $B$  has some pre-image in  $A$ :

$$\forall y \in B, \exists x \in A \text{ s.t. } f(x) = y.$$

## 4 Bijections, Inverses, and Composition

A function can be neither injective nor surjective, or it can be both at once:

$$\text{one-to-one} \cap \text{onto} = \mathbf{bijection}.$$

The inverse  $f^{-1}(y)$  is well-defined precisely when  $f$  is a bijection. For  $f : A \rightarrow B$  a bijection:

$$f(f^{-1}(y)) = y \quad (y \in B), \quad f^{-1}(f(x)) = x \quad (x \in A).$$

### 4.1 Composition

The composition of  $f$  and  $g$  is written  $(f \circ g)(x) = f(g(x))$ .

If  $g : A \rightarrow B$  and  $f : B \rightarrow C$ , then  $f \circ g : A \rightarrow C$ . In general  $f \circ g \neq g \circ f$ .

**Example.** Let  $f(x) = \sqrt{x}$  over  $\mathbb{N}$  and  $g(x) = x^3$  over  $\mathbb{Z}$ . Evaluating  $f(g(x))$  at  $x = -1$ :  $g(-1) = -1$ , then  $f(-1)$ . (As recorded in the original notes; note that  $f = \sqrt{x}$  is only defined for nonnegative inputs, so this step is worth double-checking against the lecture.)

## 5 Countably Infinite Sets

### 5.1 $|\mathbb{N}| = |\mathbb{Z}|$

Define  $f : \mathbb{Z} \rightarrow \mathbb{N}$  by

$$f(i) = \begin{cases} 2i & \text{if } i \geq 0 \\ -(2i + 1) & \text{if } i < 0 \end{cases}$$

| $i$      | $f(i)$   |
|----------|----------|
| -3       | 5        |
| -2       | 3        |
| -1       | 1        |
| 0        | 0        |
| 1        | 2        |
| 2        | 4        |
| $\vdots$ | $\vdots$ |

$f$  is a **bijection** (one-to-one and onto). If a bijection can be built between two sets, they have the same cardinality; if one side is  $\mathbb{N}$ , both sets are **countably infinite**.

## 5.2 $\mathbb{N}^2$ is countably infinite

Let  $\mathbb{N}^2 = \mathbb{N} \times \mathbb{N} = \{(0, 0), (0, 1), (1, 0), (1, 1), \dots\}$ . Is this countable, and if so, countably infinite?

Define the pairing function  $f : \mathbb{N}^2 \rightarrow \mathbb{N}$  (positive integers) by

$$f(a, b) = 2^a(2b + 1).$$

(A bijection  $\mathbb{N}^2 \rightarrow \mathbb{N}_{\geq 1} \rightarrow \mathbb{N}$  just shifts by  $f(i) = i - 1$  at the end, to match whichever convention  $\mathbb{N}$  starts at.)

**Claim.**  $f$  is a bijection.

*Proof.* ( **$f$  is one-to-one.**) Suppose  $f(a_1, b_1) = f(a_2, b_2)$  with  $(a_1, b_1) \neq (a_2, b_2)$ , i.e.

$$2^{a_1}(2b_1 + 1) = 2^{a_2}(2b_2 + 1).$$

*Case 1:*  $a_1 = a_2$ . Then  $2^{a_1}[(2b_1 + 1) - (2b_2 + 1)] = 0$ . Since  $2^{a_1} \neq 0$ , this forces  $b_1 = b_2$ , contradicting  $(a_1, b_1) \neq (a_2, b_2)$ .

*Case 2:*  $a_1 \neq a_2$ , WLOG  $a_2 > a_1$ . Dividing both sides by  $2^{a_1}$ :

$$(2b_1 + 1) = 2^{a_2 - a_1}(2b_2 + 1).$$

The left side is *odd*; the right side is *even* (since  $a_2 - a_1 \geq 1$ ). Contradiction.

So no two distinct inputs share an output, i.e.  $f$  is injective.

( **$f$  is onto.**) We show every  $n \geq 1$  can be written as  $n = 2^a(2b + 1)$ , by strong induction on  $n$ .

*Base case:*  $n = 1 = 2^0(2 \cdot 0 + 1)$ , so  $a = 0$ ,  $b = 0$ .

*Inductive step:* assume every  $i < n$  can be written in this form.

- If  $n$  is odd, write  $n = 2k + 1 = 2^0(2k + 1)$ , so  $a = 0$ ,  $b = k$ .
- If  $n$  is even, write  $n = 2k$ . Since  $k < n$ , by the inductive hypothesis  $k = 2^a(2b + 1)$  for some  $a, b$ , so

$$n = 2 \cdot 2^a(2b + 1) = 2^{a+1}(2b + 1).$$

Either way  $n$  has the required form, so every  $n \in \mathbb{N}$  is hit, i.e.  $f$  is surjective. ■

Since  $f$  is both injective and surjective, it is a bijection, so  $\mathbb{N} \times \mathbb{N}$  is countably infinite.

## Practice Questions

1. Let  $A = \{1, 2\}$  and  $B = \{x, y, z\}$ . List  $A \times B$  and  $B \times A$ , and verify that  $|A \times B| = |A| \cdot |B|$ .
2. Let  $f : \{p, q, r\} \rightarrow \{1, 2, 3, 4\}$  be given by  $f(p) = 1$ ,  $f(q) = 4$ ,  $f(r) = 1$ . State the domain, codomain, and range of  $f$ . Is  $f$  injective? Is it surjective?
3. Give an example of a function  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  that is injective but not surjective, and one that is surjective but not injective. Justify each.
4. Let  $f(x) = 2x + 1$  and  $g(x) = x^2$ , both over  $\mathbb{R}$ . Compute  $(f \circ g)(x)$  and  $(g \circ f)(x)$ , and confirm that  $f \circ g \neq g \circ f$ .
5. Explain, in your own words, why constructing a bijection  $f : A \rightarrow B$  shows that  $A$  and  $B$  have the same cardinality. Then use this idea to argue that the set of even integers is countably infinite.